

Khushi Raj

Fourth year, BS-MS (Integrated) IISER Berhampur

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Research Interest

- Quantum Information Theory
- Quantum Computation
- Quantum Cryptography
- Quantum Foundations

Education

Indian Institute of Science Education and Research Berhampur

Sept 2023 – Present

- BS-MS (Physics Major)
- CPI: 8.0/10

Relevant Coursework

- **Coursework and Labs in Physics:** Quantum Mechanics I, Quantum Mechanics II, Mathematical Methods I, Mathematical Methods II, Statistical Mechanics, Thermal Physics, Electromagnetism, Classical Mechanics, Waves and Optics, Optics Lab, Basic Electronics Lab.
- **Additional Coursework:** Mathematics, Computer and Data Science, Chemistry, Biology, Earth and Environmental Science, and Humanities and Social Science.
- **Other Certification:** Machine Learning Specialization by Andrew Ng (Coursera)

Research Experience

Prof. Ritabrata Sengupta (IISER Berhampur)

May 2026 – Present

- Studied the theory of Mutually Unbiased Bases (MUBs) through the construction based on Weyl operators and maximal commuting classes of unitary operators.
- Constructed complete sets of MUBs in prime and prime-power dimensions, and analyzed why this construction does not extend directly to composite dimensions.
- Constructed and analyzed examples of weakly and strongly mutually unbiased bases, and studied MUBs using maximal abelian subspaces of matrix algebras.

Prof. Utpal Roy (IIT Patna)

Jun 2025 – July 2025

- Studied the foundations of quantum computation and quantum information, covering key concepts including quantum states and measurements, entanglement, Bell states, the CHSH game, quantum teleportation, superdense coding, and POVMs.
- Learned Qiskit and implemented various quantum protocols to gain hands-on experience with quantum programming.
- Explored quantum machine learning, focusing on amplitude encoding techniques.
- Conducted an in-depth study of research papers on Quantum Key Distribution (QKD).
- Presented key findings and insights from the literature through multiple technical presentations.

Seminars Attended

Mathematics for Data Security (IIT Madras)

May 2025

- Introduction to Lattice-Based Cryptography
- Decentralization, Distributed Trust, and Blockchain
- Distributed Graph Algorithms and Congestion
- Factoring, RSA Security, and Lattice Attacks

Skills

Programming Languages: Python, MATLAB, JavaScript, Qiskit

Tools and Technologies: Git and Github, Jupyter, SQL, MongoDB, PHP

Languages: English, Hindi